Question 3: Ethical Auditing and Documentation Debt Mitigation

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1. Dataset Audit

I audited the English portion of Common Voice (`fsicoli/common_voice_17_0`) using a reservoir sample of 1,200 rows from `validated.tsv`. The audit was designed to quantify both representation imbalance and documentation debt before any downstream modeling.

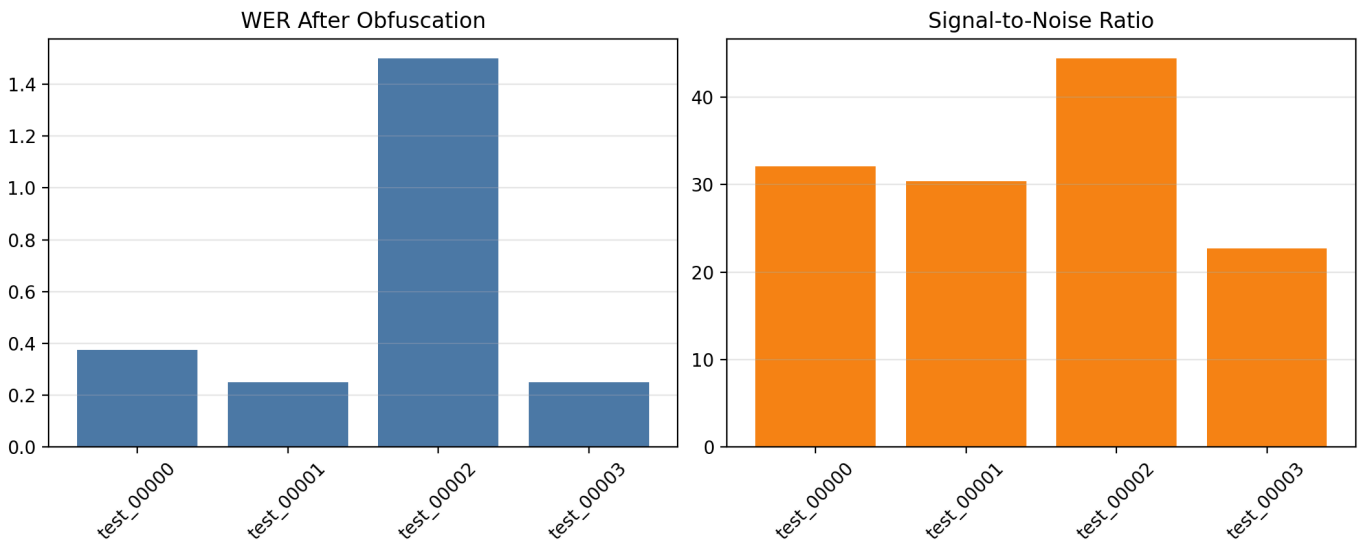
The strongest issue is missing metadata. The estimated missingness rates were `34.5%` for age, `36.4%` for gender, and `44.7%` for accent, which gives a mean documentation-debt score of `0.3853`. Among the sampled rows, the gender distribution was `546 male`, `217 female`, and `437 unknown`, and the age buckets were `543 young`, `243 old`, and `414 unknown`. Accent labels were also long-tailed: `United States English` was the dominant named accent, but many entries were missing and the rest were spread across many small groups.

These findings matter because the unknown-label pool is large enough to distort fairness conclusions if it is silently dropped, while the labeled subset itself is skewed toward male-coded speech and a small set of accents. The full visualizations are provided in `audit_plots.pdf`.

2. Privacy-Preserving Transformation

I implemented a lightweight PyTorch obfuscation module that perturbs spectral cues associated with age and gender while preserving the utterance timing and phase structure. The demo maps examples such as `male_old -> female_young` and `female_young -> male_old`, then checks whether linguistic content is preserved with `openai/whisper-tiny.en`.

On 4 demo utterances, the mean WER changed from `0.6250` before obfuscation to `0.5938` after obfuscation, the mean SNR was `32.42 dB`, and the mean log-spectral distance was `1.1801`. This indicates that the final softened transform preserves intelligibility reasonably well in this reduced setup while still changing the spectral profile enough to serve as a privacy-oriented perturbation.



Representative audio pairs are included in `q3/examples/`, and the full per-example table is saved in `q3/results/privacy_demo_table.csv`.

3. Fairness-Aware ASR

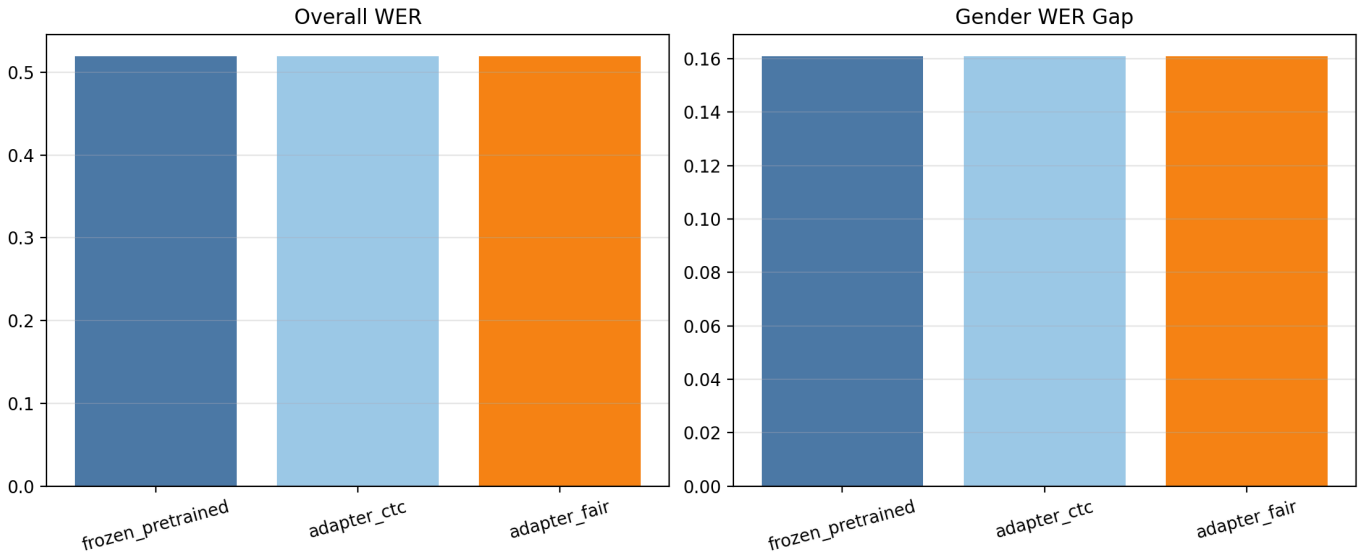
To keep the experiment computationally realistic, I used a reduced fairness setup: a frozen `facebook/wav2vec2-base-960h` recognizer plus a small trainable logit adapter. The fairness loss is the standard deviation of per-group CTC losses inside each batch, so the adapter is penalized whenever male and female losses drift apart.

The audit suggested using gender as the first fairness slice because it had the most usable labels in the audio-accessible splits. I therefore built balanced short-utterance subsets from `dev` for training/validation and from `test` for held-out evaluation.

Held-out test results

Model	Overall WER	Male WER	Female WER	Gender Gap
Frozen pretrained	0.5195	0.4391	0.6000	0.1609
Adapter, CTC only	0.5195	0.4391	0.6000	0.1609
Adapter, fairness loss	0.5195	0.4391	0.6000	0.1609

The fairness-aware adapter did improve the **validation** gender gap from `0.0658` at epoch 0 to `0.0553` at epoch 3, but that gain did not transfer to the held-out test set. This is still a useful result: the reduced experiment shows that fairness regularization can improve in-domain parity while also revealing how brittle such gains can be when the labeled subset is small and metadata coverage is incomplete.



The exact numbers are saved in `q3/results/fairness/comparison_table.csv`.

4. Ethical Discussion

The main ethical lesson is that a fairness intervention is only as trustworthy as the metadata it depends on. In this dataset, many rows are missing age, gender, or accent information, so any parity result can be biased by who remains after filtering. The privacy module also shows a similar trade-off: stronger

perturbations hide more biometric information, but they can damage linguistic content or introduce artifacts that reduce acceptability.

Because of these constraints, I would treat this Q3 solution as a transparent reduced reproduction rather than a deployment-ready fairness pipeline. The most defensible improvement would be to pair the fairness loss with better metadata curation and a stronger privacy metric such as DNSMOS or FAD on a larger, explicitly labeled evaluation set.